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PROVISIONS FOR GREATER REUSE OF STEEL STRUCTURES

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## Case studies for the implementation of reclaimed elements

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Authors: **Viorel Ungureanu, Raluca Buzatu, Daniel Muntean**  
Politehnica University Timișoara

**Petr Hradil, Sirje Vares**  
VTT Technical Research Centre of Finland

## Executive summary

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This report presents theoretical studies of building design from reclaimed elements to declare environmental and economic benefits of reused elements in the scope of RFCS project PROGRESS “Provisions for greater reuse of steel structures”.

The end-of-life qualities of construction products play an important role, considering the circular economy, as buildings have a long intended life-span and require a significant amount of material resources. In order to maintain circularity and to guarantee a limited impact, these resources have to be kept in-use through re-use, reclaim or recycling.

The report will cover the following issue:

- four different theoretical case studies presenting a comparative environmental and economic impact of the same steel building when the structure is build reusing an existing steel structure, using reclaimed elements or using new construction materials.

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## 1. Introduction

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The high environmental impact of the construction sector has determined governments and companies in Europe to find alternative solutions towards lowering greenhouse gas emissions (GHG) and the carbon footprint of buildings, the issue of climate change being a vexed question for almost three decades. In different countries, the environmental impact of the building's life cycle has been approached with slightly different emphasis. The most advanced methods are in the Netherlands, Germany, Sweden, Norway and France who have already implemented a series of internal norms and created benchmarks in sustainable development [22]. Different countries use different means of control emissions, for example, in the Dutch method LCA calculation results in the cost of environmental impacts (from using 11 different indicators), in France, the lower emissions allow you more building rights. Councils are more likely to give planning permission to greener buildings to meet government CO<sub>2</sub> targets [22]. There are also other differences in the methods, for example European Level(s) method includes office and residential buildings, while the Dutch method covers also infrastructure construction. Finnish criteria for low-carbon construction, drawn up by the Ministry of the Environment, will first be piloted in public construction projects. The goal is that carbon footprint for the whole building life cycle will be regulated in Finland by legislation, by the mid-2020s. Limit approach claimed to be taken into use in France by 2020 but in Sweden and Norway assessment of GHG emission will be mandatory. Despite the method used, the main goal is to cut high GHGs from building sector and for that, all possible means could be used.

A sustainable path should not target “green materials” only, but oppositely, it should focus on the most present structural materials, such as steel and concrete [22]. In particular, steel is preferred due to its ease of design, speed of construction and lower costs of the construction project (faster construction time will reduce the period of financing and also the buildings will be operational sooner, granting an accelerated return on investment), the capacity to provide space efficiency (using steel floor beams with large openings supports an additional floor every eight to nine storeys when compared to traditional construction materials floors), light weight (which translates into a lighter building with a diminished required foundation area). To all the above advantages, a key benefit is added, that of easy recycling of steel which can be taken even further as re-use of steel can discard most of the negative environmental impacts generated in the production process of the steel elements [22].

The maximization of structural system reuse and recycling, along with the development of new efficient production systems, the extension of the life of structures and the minimization of material consumption in new structures is also a design methodology considered so far in the sustainable approach of construction sector [22]. The end-of-life qualities of construction products play an important role, considering the circular economy, as buildings have a long intended life-span and require a significant amount of material resources. In order to maintain circularity, these resources have to be kept in-use through re-use, reclaim or recycling. The end-of-life credentials of structural materials varies notable due to their inherent properties and the final structural form, therefore the use of steel is a viable solution towards meeting circular economy criteria.

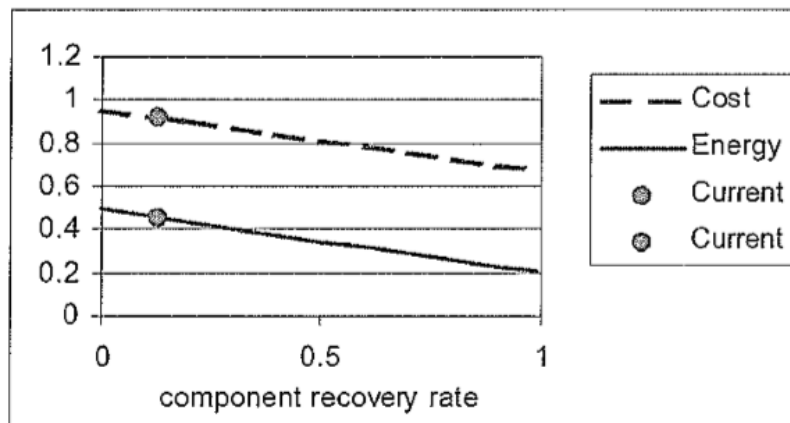
### 1.1 Recycling and reuse of steel

Recycling has been a standard of sustainable building for decades, reducing waste disposal, minimizing the use of primary material and so, inherently minimizing energy use, land used for mining (which leads intrinsically to preserving land for other users) and conserving natural

resources, allowing by this a lower embedded carbon of building materials. Considering some Environment facts, it can be mentioned [22]:

- In 2007, global steel production stood at 1.34 billion tonnes; producing one tonne of steel can release up to one tonne of carbon dioxide.
- Recycling 1 ton of steel prevents the need for about 900 kilograms of iron ore, 635 kilograms of coal, 363 kilograms of petroleum coke, and 204 kilograms of lime. These reductions in mining burden also reduced impacts to the land, air and water.
- Recycling 1 ton of steel saves the energy equivalent of about 380 liters of oil. Therefore, reusing steel will save so much more energy.
- Compared to producing steel from raw iron ore, recycling steel reduces air pollution by about 86% and reduces water pollution by around 76%. In particular, producing 1 ton of steel from raw iron ore produces around 245 kilograms of solid wastes and 19 kilograms of air pollutants.

The energy savings from steel and iron recycling (compared to production using primary materials) are about 72% [22] yet, recycling process requires a large amount of energy for transport, process and reassembling. Reusing is a step forward to sustainable building than recycling, as reusing saves the energy that comes with the re-manufacturing of the products. Figure 1 [22] shows how the system energy and cost consumption changes as the reuse rate of the recovered steel sections grows, for a total recovery rate (components and materials) of 99% of the steel sections at the end-of-life of a structure:



*Fig. 1: Shift from recycling to reuse – Normalised cost and energy as function of component recovery rate (total recovery rate = 0.99) [22]*

The energy savings are doubled when shifting from recycling to reuse. There are strong environmental and economic encouragements that favour a shift from recycling to reusing structural steel sections, but a constrained reuse yield can fully reshape the environmental and economic performance of reuse [22]. Research [22] confirms that there are significant barriers to steel reuse such as deconstruction costs and challenges in recovering sections from buildings, lack of standardisation for reused construction elements, concerns about the quality of the reused steel, the lack of an infrastructure of a supply chain between users and re-users, reluctance to change current practices, disruptions causing delays to the project timeline.

An Eurofer survey [22] performed in 2001 and 2012 (with further additions in 2013) reveals that there is an improvement in the overall recycling and reuse rate of the steel construction products in the last two decades, the results showing an increment from 93% (in 2001) to 96% (in 2012) for the reused and recycled (combined) steel construction elements. However, reuse rate estimates depreciated between 2001 and 2012, showing that on average, only 5% of steel construction products were reused in 2012 in comparison with 2001 survey when 8% of these products were concluded to be reused, revealing that reusing structural steel is an underutilized process.

*Table 1: Comparison between the 2000 and the 2012 survey results (including further additions from 2013) regarding the reuse and the recycling rates of steel construction products [22]*

| Product                                                         | %Reused |      | %Recycled |      | %Lost |      |
|-----------------------------------------------------------------|---------|------|-----------|------|-------|------|
|                                                                 | 2001    | 2012 | 2001      | 2012 | 2001  | 2012 |
| Heavy structural sections / tubes                               | 12      | 7    | 87        | 93   | 1     | 0    |
| Rebar (in concrete superstructure)                              | 2       | 0    | 90        | 98   | 8     | 2    |
| Light structural steel (e.g. galvanized purlins, supports etc.) | 10      | 5    | 89        | 93   | 1     | 2    |
| Profiled sheet cladding (roof/ facade)                          | 15      | 10   | 79        | 89   | 6     | 1    |
| Other non-structural steel                                      | 2       | 4    | 85        | 96   | 13    | 1    |
| Average (across all products)                                   | 8       | 5    | 85        | 91   | 7     | 4    |

## 1.2 Overcoming barriers to steel reuse

To encourage reusing steel in constructions one has to soften the barriers that restrain this. A survey along with an interview [22] among the main actors involved in the reuse chain (large firms from the construction industry, architects, structural engineers, main contractors, fabricators stockist, demolition contractors) reported a series of barriers as being the top impediments in reusing structural steel:

*Table 2: Top two barrier for each actor, including the scores for each barrier [22]*

| Actor                  | Top barrier                            | Score | Second barrier                         | Score |
|------------------------|----------------------------------------|-------|----------------------------------------|-------|
| Architects and clients | Trust / lack of communication          | 1.40  | Old / new perception                   | 1.25  |
| Structural engineers   | Design for deconstruction              | 1.35  | Availability / dimensions              | 1.15  |
| Main contractors       | Availability / dimensions              | 1.26  | Uncommon practice                      | 1.04  |
| Fabricators            | Quality / certification / traceability | 1.46  | Uncommon practice                      | 1.17  |
| Stockists              | Old / new perception                   | 4.51  | Quality / certification / traceability | 1.46  |
| Demolition contractors | Design for deconstruction              | 2.70  | Programme                              | 1.45  |

Improving the chances for structural steel reuse starts when a building is designed, not when it reaches the end of its life. To guarantee a good yield of reusing, first step forward is to design having in mind the full life-cycle of the building, which means designing for adaptability, durability and with materials suitable for reutilisation. Many building materials are not reusable or they are interconnected in ways that makes difficult for separation and reuse even if, theoretically, they are reusable [22]. Design for deconstruction provides the opportunity to reduce the cost associated with dismantling and preparing for reuse that are faced nowadays.

Practical applications of design for deconstruction would scale down deconstruction costs and, therefore, the reduction in total life cycle cost of reuse [22]. As the chance of being able to remove a component without damage depends notably on the installation method and building typology [22], design for deconstruction grows into a relevant barrier to overcome in order to improve the technical reusability of steel. A considerable higher rate of reusing structural steel could be achieved by implementing design for deconstruction for steel-framed industrial buildings, as industrial buildings have a short intended lifespan in the last decades (avg. 25 years) and the annual steel consumption in such buildings expanded from 2 to 3 Mt between 1992 and 2008 (according with the ECCS statistics for the 17 largest European countries) [22].

A design for reuse in the planning phase of a structure and / or designing with reused structural steel needs a clear method in evaluating the performance of the to-be-reused steel elements regardless the way in which steel is reused: in-situ or in a different location, reusing constituent products, structural components or reusing the entire primary structure [22]. A clear path in classification of the reused steel would surmount another barrier in the reuse process as it is the lack of standardisation or the concerns about quality and certification of potentially to-be-reused steel construction elements. Ungureanu et al. (2019) proposed an approach for the integration of reclaimed steel in different reuse scenarios:

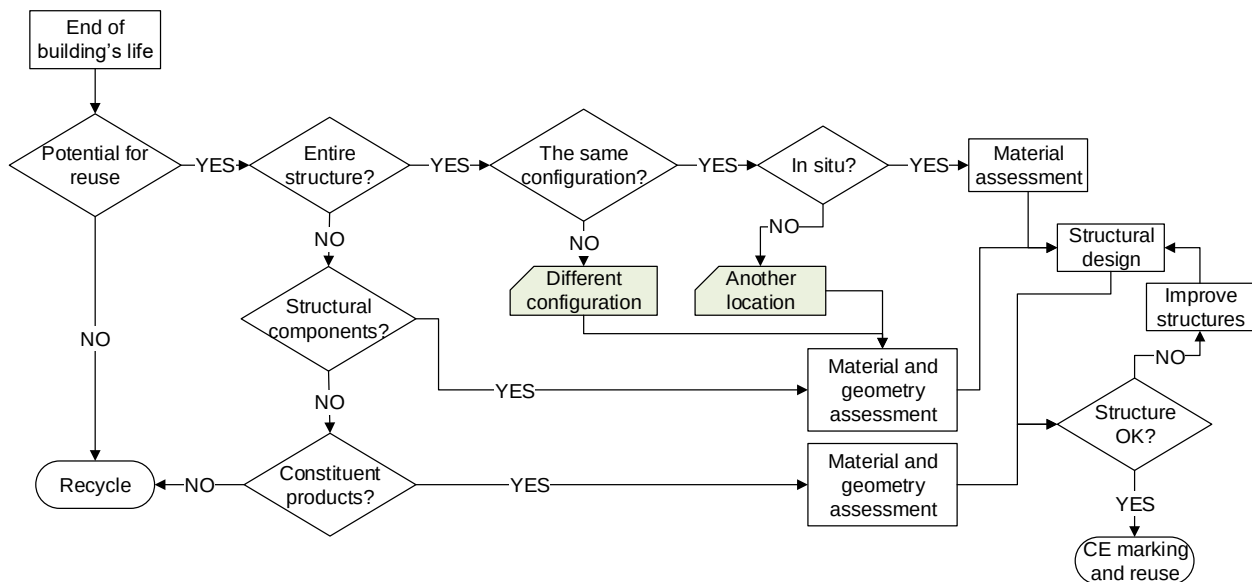


Fig. 2: Integration of reclaimed steel in different reuse scenarios [22]

The model of the interconnection between main actors involved in the reuse chain suggested by Cullen et al. (2016) would help the lack of communication / the lack of an infrastructure of a supply chain between users and re-users supporting a higher rate of reuse steel construction products.

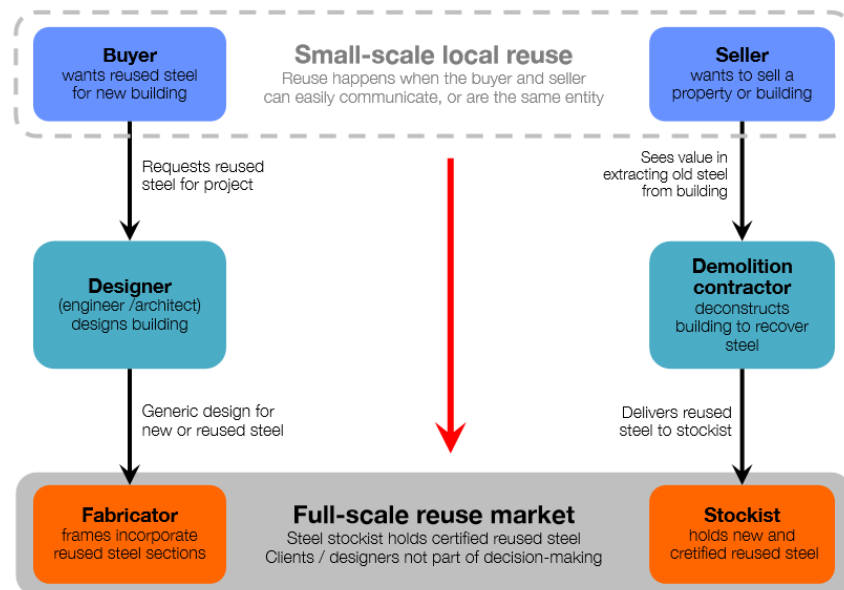


Fig. 3: Hypothetical link between main actors involved in the reuse chain [22]

A strong integration in the supply chain assures a practical and cost effective reuse of steel, as shown in the work of Gorgolewski et al. [15], in which a collection of successful reuse projects is presented. The reuse case studies, documented in Canada, in which constituent products and structural components (roof trusses) were reused, validated that the impediments to reuse structural steel can be surmounted in the right market conditions [22].

There were several successful reuse projects in Europe also, as noted in Deliverable D 1.1 “Factsheets on review of existing deconstruction cases” [22] where case studies of reuse steel structures around Europe have been identified and described in detail. In Deliverable D 1.2 “Report on the reuse potential of single-storey steel buildings” [22] along with the examples detailed in D1.1, other 13 examples of reuse scenarios were presented and labelled by the locational scenarios of which a steel structure can be reused (reuse in-situ, reuse on the same site, reuse in a different site). These reuse projects demonstrated that structural steel reuse is possible and it represents a viable solution in construction industry, yielding considerable benefits in terms of savings of embedded carbon in steel, time savings or cost savings.

## 2. Aim of the study case

The purpose of the study was to compare environmental and economic indicators (in terms of impacts such as GHG emissions and costs) of steel structures assessing structures made from elements using new materials along with structures made from reused steel components, quantifying the savings achievable by reusing construction materials. The study is based on an environmental impact assessment and on a life cycle cost assessment of a single-storey industrial hall erected in Romania.

For the assessment were carried out and compared six different case-scenarios:

1. Baseline scenario (case 0) where the structure was designed as a new structure made with elements from new materials (optimal design).
2. Second scenario (case 0+) where the structure was designed with elements from new materials, taking in mind the deconstruction at the end-of-life of the structure (a new structure made with elements from new materials designed for deconstruction) – **covering Deliverable 7.3 regarding Design for Deconstruction.**
3. Third scenario (case 1) regarding a relocated steel structure; the scenario considered the reuse of an existing steel structure originated in Germany and reassembled in Romania.
4. In the fourth scenario (case 2) it was weighed a steel structure made with reclaimed elements. Existing profiles for beams and columns have been identified in a storage yard in Germany deriving from other deconstructed buildings. All other components were made with new steel.
5. The fifth scenario (case 3) closeness Case 2, considering reclaimed elements such as columns and beams but also end plates for beams and columns. All other components represent new steel.
6. The last scenario (case 4) considered the reuse of an entire structure relocated from Germany. The percent of the reused steel in superstructure in this scenario is 100%.

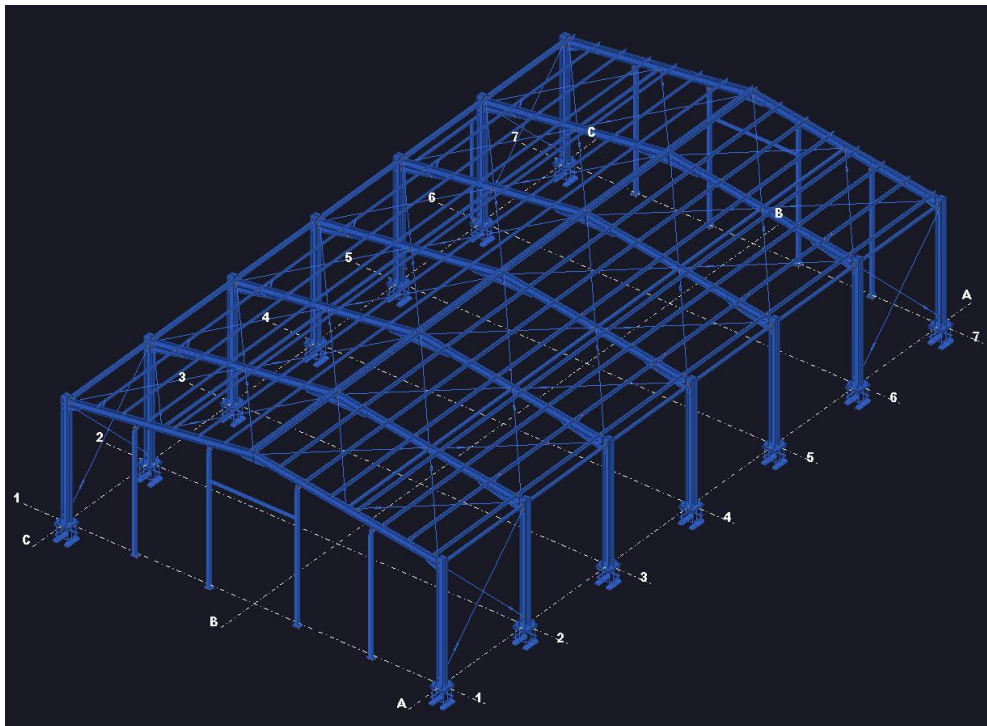
The amount of steel consumption and reused steel percent for each case scenario can be seen in the table below:

*Table 3: Steel consumption (for the superstructure) – total steel vs. reused steel*

| Case no. | Total weight [kg] | Reused steel [kg] | %    |  | Purlins [kg] |
|----------|-------------------|-------------------|------|--|--------------|
| CASE 1   | 24812.7           | 17907.1           | 72.2 |  | 2695.1       |
| CASE 2   | 27716.3           | 16795.4           | 60.6 |  | 2550.4       |
| CASE 3   | 29112.2           | 19919.7           | 68.4 |  | 2550.4       |
| CASE 4   | 27716.3           | 27716.3           | 100  |  | 2550.4       |
| CASE 0   | 23682.8           | -                 | -    |  | 2550.4       |
| CASE 0+  | 24593.4           | -                 | -    |  | 2550.4       |

## 2.1 Case 0

The study is built on a single-storey industrial hall located in Romania having a total length of 30 m, with six identical frames at 5 m spacing. The wide is 17.5 m and eaves height is 6 m. The load bearing structure is constituted of hot rolled steel sections (columns – HEA320, beams – IPE300, S355) and the envelope involves steel sandwich panels with mineral wool insulation (120 mm sandwich panels for the roof and 80 mm sandwich panels for the walls).



*Fig. 4: Baseline scenario structure*

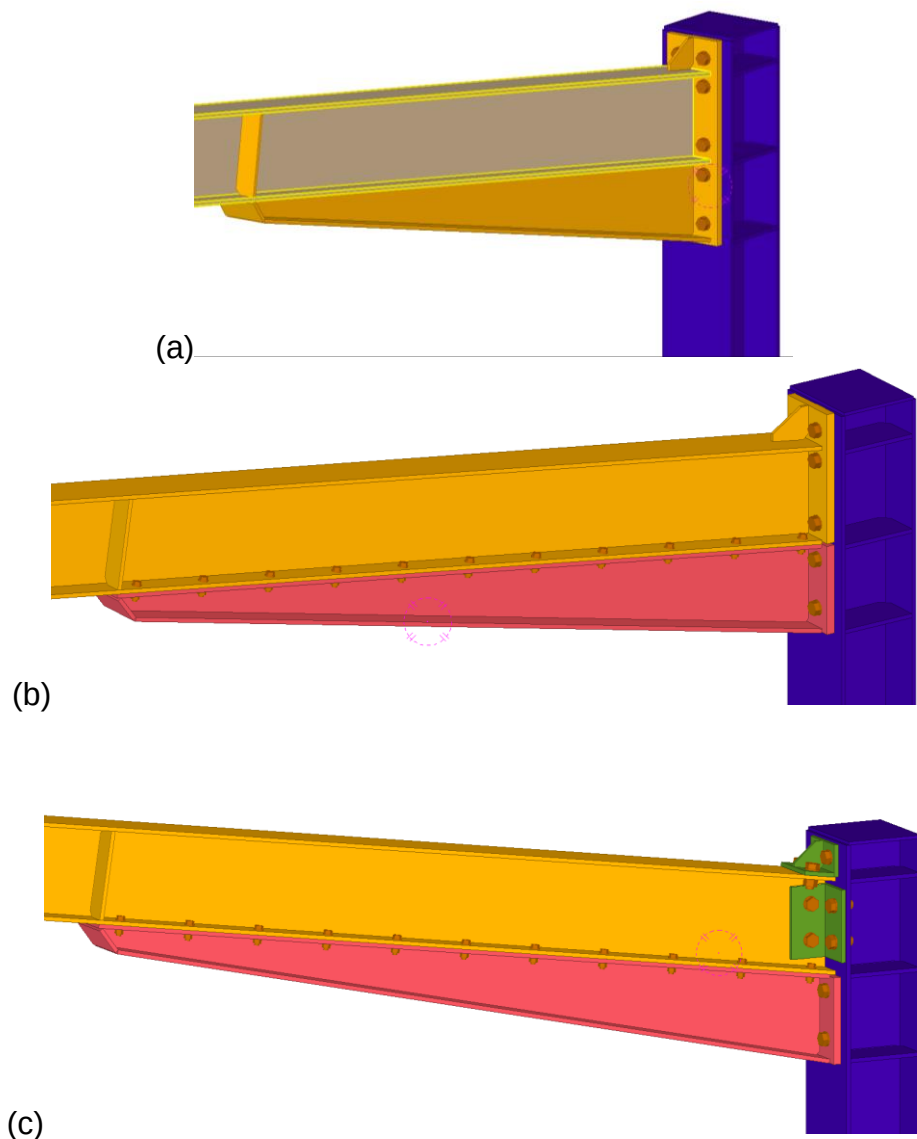
All the elements of the structure were designed as elements made from new materials. The total weight of steel consumption was 23682.8 kg + 2550.4 kg (hot rolled steel profiles + cold formed purlins).

## 2.2 Case 0+

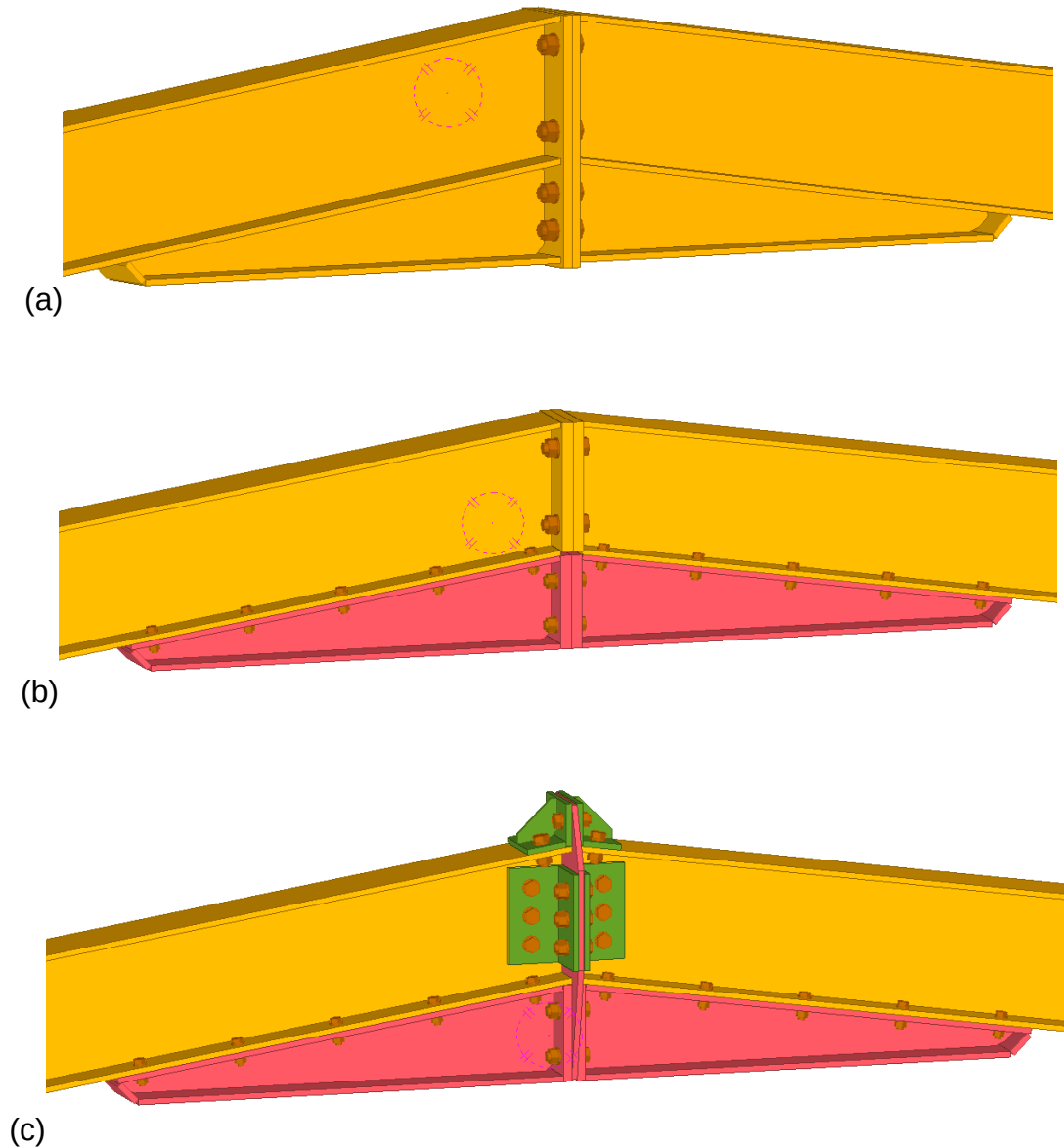
The case study is built on an industrial hall located in Romania which has the same dimensions as the one from previous case: a total length of 30 m, with six identical frames at 5 m spacing, a wide of 17.5 m and eaves height of 6 m. The load bearing structure is constituted of hot rolled steel sections (columns – HEA320, beams – IPE300, S355) and the envelope involves steel sandwich panels with mineral wool insulation (120 mm sandwich panels for the roof and 80 mm sandwich panels for the walls). All the elements of the structure were designed as elements made from new materials. **The difference from Case 0 is given by the fact that this structure was designed for deconstruction which brings an extra material consumption for the structure of almost 1t of steel. The total weight of steel consumption was 24593.4 kg + 2550.4 kg (hot rolled steel profiles + cold formed purlins). Case 0+ covers also Deliverable 7.3 regarding Design for Deconstruction.**

In order to ensure a demountable structure at the end-of-life, several alternatives for classic connections used in the case of steel portal frames were studied. The studied alternatives have similar structural performances with the classic connections. For this purpose, IDEA StatiCa software has been used. IDEA StatiCa Connection designs and checks welded or bolted connections, 2D or 3D joints, anchoring or footing of all topologies. For analysis, the Unique CBFEM method based on finite elements was used. The model considers an elastic-plastic material model for all the elements of the joint - plates, elements, bolts and welds. Pre-stressed bolts and plastic behaviour of the welds are also considered which provides clear pass / fail checks according to EC / AISC / CISC standards in a complete report with drawings.

On the following, two types of classic connections were investigated, i.e. eaves and apex. The alternatives are (1) connection with endplate and detachable haunch and (2) connection with detachable endplate and haunch using cleat angles. Figures 5 and 6 presents the classic solution and the two alternatives, for both eaves and apex connections.



*Fig. 5: Eaves connections: (a) classic solution; (b) alternative 1; (c) alternative 2*



*Fig. 6: Apex connections: (a) classic solution; (b) alternative 1; (c) alternative 2*

The design moment resistance and the initial rotational stiffness of each joint is presented in table 4 where, the similar performances for the three solutions can be seen. Table 4 presents also the necessary number of bolts/diameter/class and additional steel consumption for the three solutions. Additional steel consumption is presented at the level of endplate and additional components only, i.e. (1) weight of endplate only; (2) weight of endplate and the top flange of the haunch; (3) weight of cleat angles and the top flange and endplate of the haunch.

Table 4: Structural performances of the designed-for-deconstruction connections

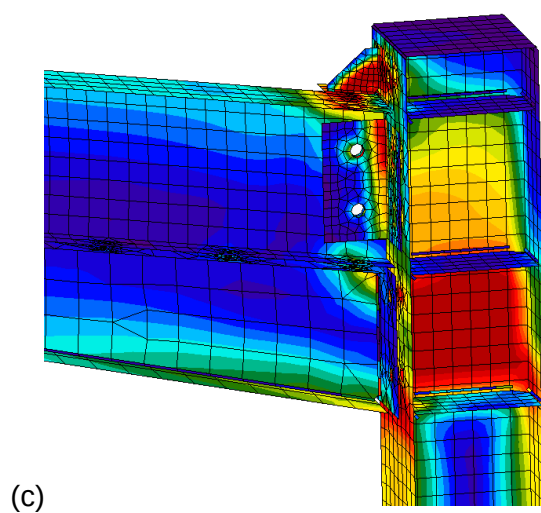
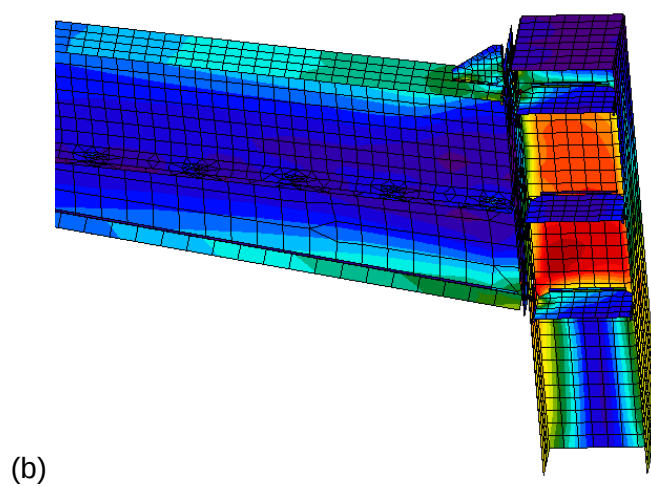
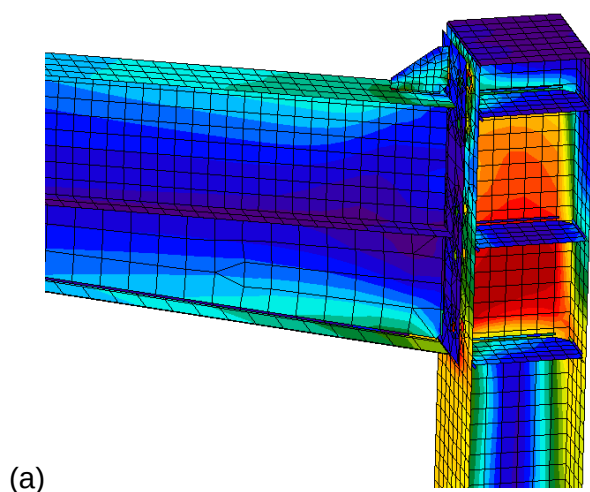
| Description      |     |                                                                                      | $M_{Ed}$<br>[kNm] | $M_{Rd}$<br>[kNm] | $S_{j,ini}$ | Consumption         |       |           |       |
|------------------|-----|--------------------------------------------------------------------------------------|-------------------|-------------------|-------------|---------------------|-------|-----------|-------|
|                  |     |                                                                                      |                   |                   |             | Steel<br>[kg]       | Bolts |           |       |
|                  |     |                                                                                      |                   |                   |             |                     | No.   | d<br>(mm) | class |
| Eaves Connection | 1.1 | Classic eaves connection                                                             | -119.75           | -182.2            | 33.2        | 66.4 <sup>(1)</sup> | 10    | 20        | 8.8   |
|                  | 1.2 | Eaves connection with endplate and detachable haunch                                 | -119.75           | -183.3            | 24.2        | 85.1 <sup>(2)</sup> | 10    | 20        | 8.8   |
|                  |     |                                                                                      |                   |                   |             |                     | 22    | 12        | 8.8   |
|                  | 1.3 | Eaves connection with detachable endplate and haunch using cleat angles (no welding) | -119.75           | -138.2            | 46.3        | 84 <sup>(3)</sup>   | 4     | 20        | 10.9  |
|                  |     |                                                                                      |                   |                   |             |                     | 10    | 20        | 8.8   |
|                  |     |                                                                                      |                   |                   |             |                     | 22    | 12        | 8.8   |
| Apex Connection  | 2.1 | Classic apex connection                                                              | -60               | -75.9             | Infinite    | 35.6 <sup>(1)</sup> | 8     | 20        | 8.8   |
|                  | 2.2 | Apex connection with endplate and detachable haunch                                  | -60               | -77.8             | Infinite    | 54.1 <sup>(2)</sup> | 4     | 20        | 8.8   |
|                  |     |                                                                                      |                   |                   |             |                     | 4     | 16        | 8.8   |
|                  |     |                                                                                      |                   |                   |             |                     | 20    | 12        | 8.8   |
|                  | 2.3 | Apex connection with detachable endplate and haunch using cleat angles (no welding)  | -60               | -77.8             | 15.5        | 62.1 <sup>(3)</sup> | 6     | 16        | 10.9  |
|                  |     |                                                                                      |                   |                   |             |                     | 16    | 16        | 8.8   |
|                  |     |                                                                                      |                   |                   |             |                     | 20    | 12        | 8.8   |

(1) weight of endplate only

(2) weight of endplate and the top flange of the haunch

(3) weight of cleat angles and the top flange and endplate of the haunch

Figures 7 and 8 present the stress distribution on the joint components.



*Fig. 7: Stress distribution for the eaves connections: (a) classic solution; (b) alternative 1; (c) alternative 2*

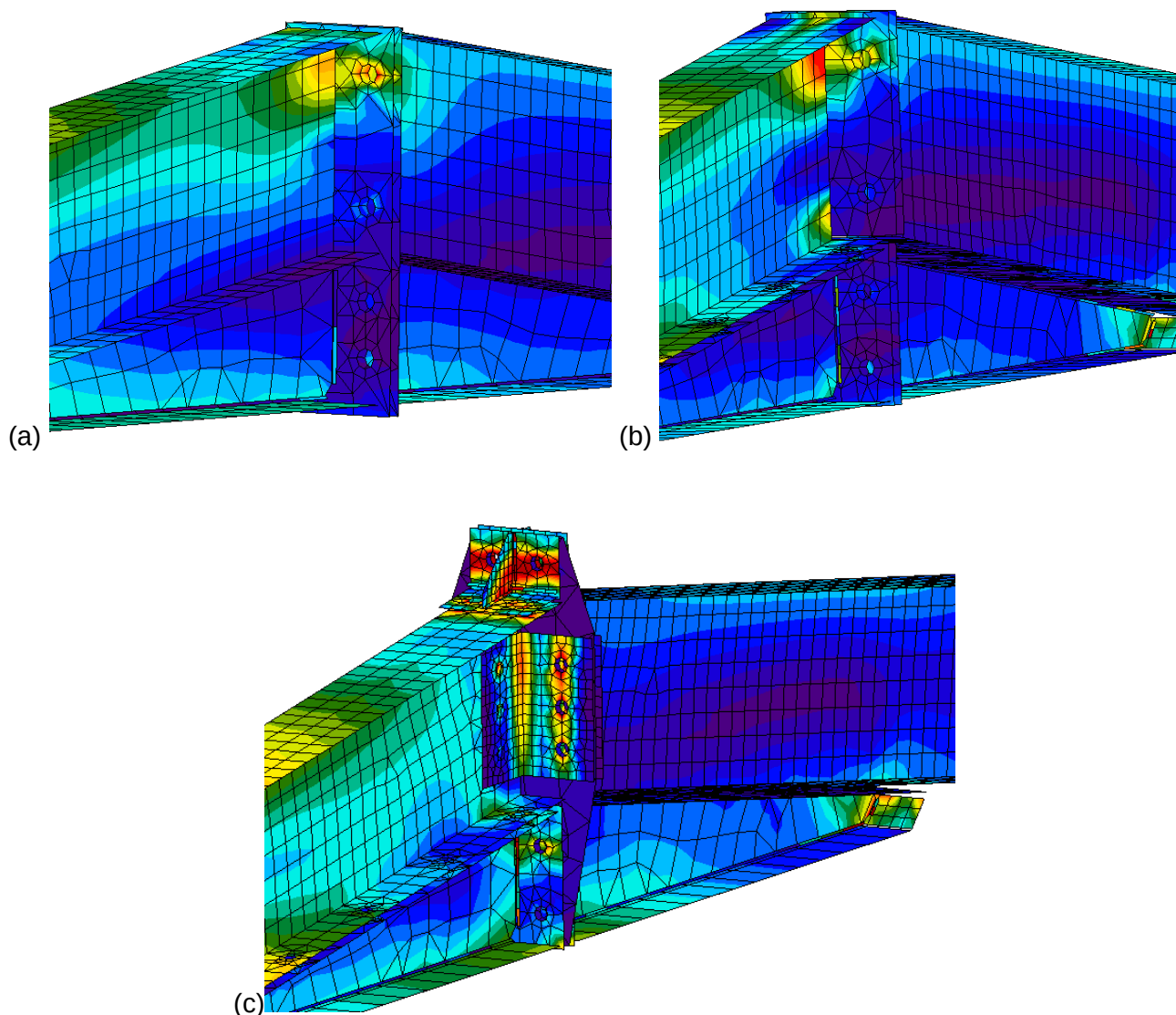


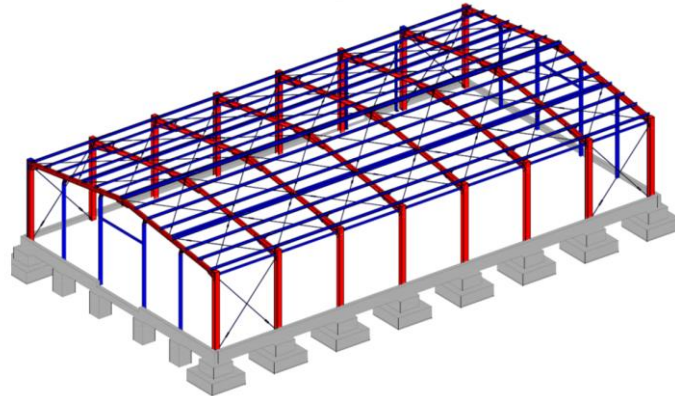
Fig. 8: Stress distribution for the apex connections: (a) classic solution;  
 (b) alternative 1; (c) alternative 2

The results presented here are available for a given practical situation but can be extended for various typologies. Present study does not include optimisation of all components. In this study, in the Case 0+, the analysis included the alternative 1 for connections, as presented in figures 5 and 6 (eaves and appex connections with endplates and detachable haunches).

## 2.3 Case 1

The structure is originated in Germany and it had (initial dimensions) 17.5 m span, 35 m length (7 bays of 5 m each) and 6 m eaves height. The beams were IPE 360 steel profile with variable cross-section at the beam-to-column connection and the columns were HEA 300. The steel grade is S355. In order to be relocated in Romania, the structure had to bear higher loadings than in Germany (characteristic values: snow – 1.5 kN/m<sup>2</sup>, wind – 0.39 kN/m<sup>2</sup>). To achieve this condition, it was concluded that only the portal frames of the structure can be reused either with a smaller bay than the initial one or by reinforcing the frames to be able to carry the new loads. Due to the highly price of the second option, it was decided to build the structure with a bay of 0.5 m smaller than the

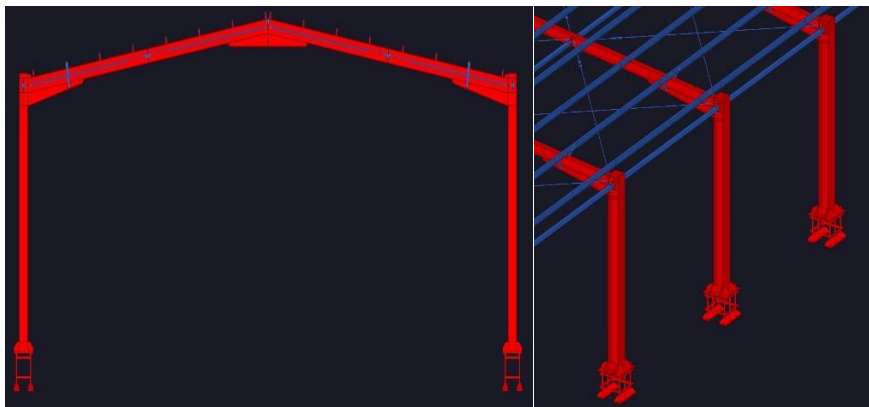
original one. In the end, the structure to be rebuilt in Romania has 31.5 m length, 7 bays of 4.5 m and 17.5 m wide.



*Fig. 9: The steel structure rebuilt in Romania (reused steel in red, new steel in blue)*

The existing roof, the wall bracing systems and the longitudinal beams were not strong enough to conform to the seismic requirements, which constrained their manufacture of new steel. In the end, the total weight of the steel structure is 27507.8 kg + 2695.1 kg (hot rolled steel profiles + cold formed purlins), from which, 19717.97 kg represent reused steel.

The existing envelope consisting of 80 mm steel sandwich panels with mineral wool insulation was reused for the structure erected in Romania but, in order to comply with the U-values existing in Case 0, an additional layer of 60 mm sandwich panels (new elements) was added to the entire envelope.

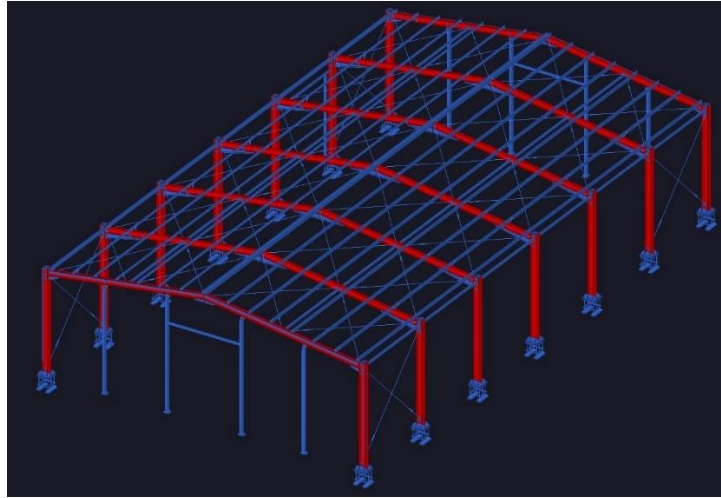


*Fig. 10: Transversal frame of the steel structure rebuilt in Romania (left) and eaves detail (right) - reused steel in red, new steel in blue*

## 2.4 Case 2

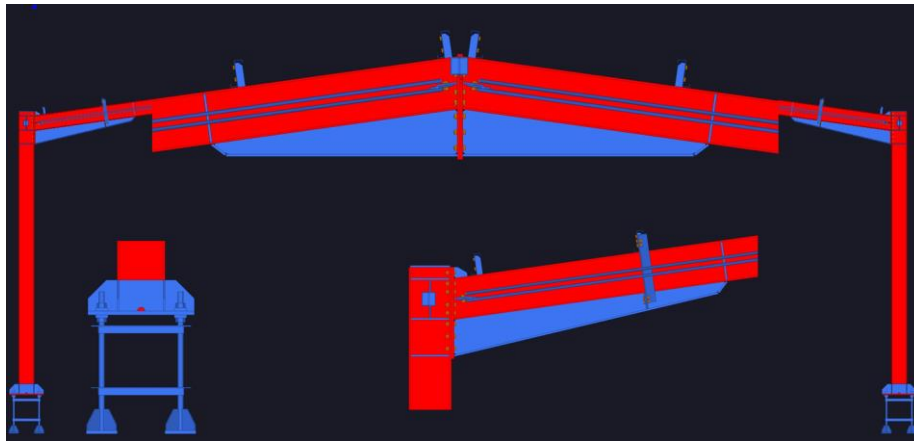
Existing profiles (as individual members) for beams and columns have been identified in a stockyard in Germany deriving from other deconstructed buildings. The beams are IPE 360 steel profile and the columns are HEA 400. The steel grade is S275. All other components represent new steel. As in Case 0, the structure has a total length of 30 m, with six identical frames at 5 m spacing, a 17.5 m wide and height at eaves of 6 m. The total weight of the steel structure is

27716.3 kg + 2550.4 kg (hot rolled steel profiles + cold formed purlins), from which, 16795.4 kg represent reused steel.



*Fig. 11: The steel structure considered in case 2 (reused steel in red, new steel in blue)*

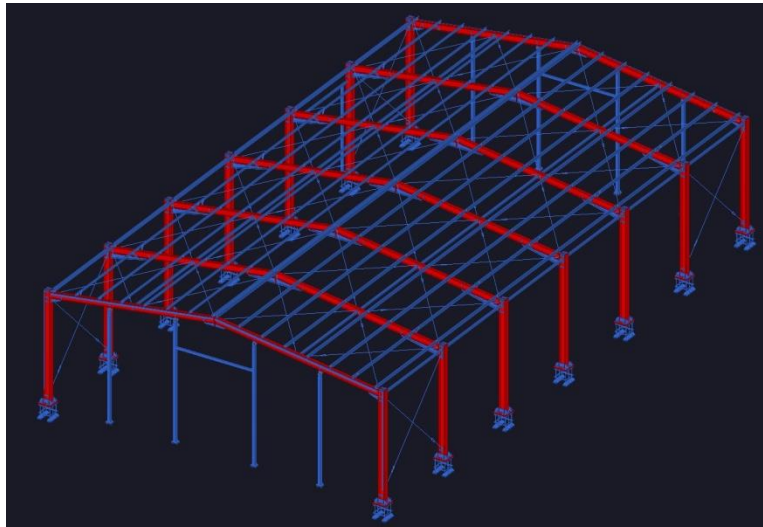
The envelope consists, as in Case 1, in 80 mm steel sandwich panels with mineral wool insulation (reused elements) plus 60 mm mineral wool sandwich panels (new elements) to respect with the U-values existing in Case 0.



*Fig. 12: Transversal frame of the steel structure in Case 2, with details of apex connection (upper center), column-base connection (bottom left) and eaves connection (bottom center) - reused steel in red, new steel in blue*

## 2.5 Case 3

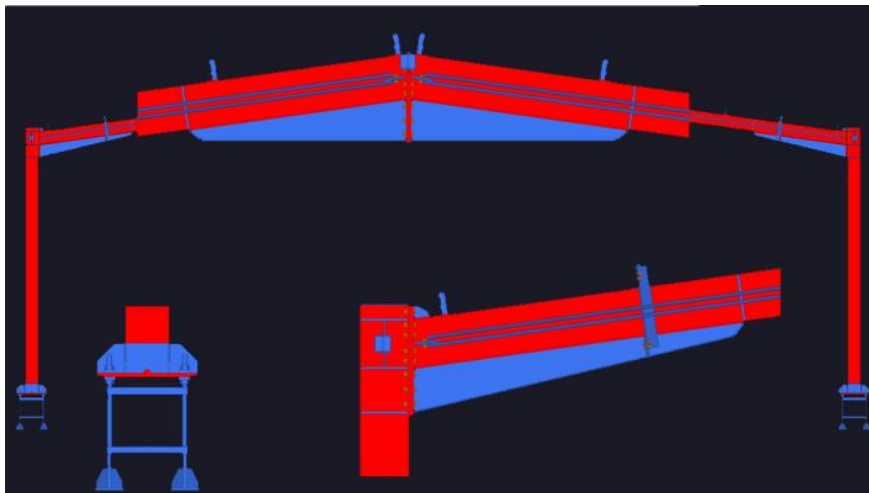
Individual members of steel profiles for beams and columns have been identified in a stockyard in Germany along with end plates for beams and columns. The beams are IPE 360 steel profile, the columns are HEA 400 and the end plates are 30 mm thick. The steel grade is S275. All other components of the steel structure represent new steel. The dimensions of the steel structure are the same as in Case 0 and in the Case 2.



*Fig. 13: The steel structure considered in case 3 – reused hot rolled steel profiles and end plates (reused steel in red, new steel in blue)*

The total weight of the steel structure is 29112.2 kg + 2550.4 kg (hot rolled steel profiles + cold formed purlins), from which, 19919.7 kg represent reused steel.

The envelope has the same composition as in cases 1 and 2: 80 mm reused steel sandwich panels + 60 mm new steel sandwich panels, both with mineral wool insulation.



*Fig. 14: Transversal frame of the steel structure in Case 3, with details of apex connection (upper center), column-base connection (bottom left) and eaves connection (bottom center) - reused steel in red, new steel in blue*

## 2.6 Case 4

The structure is originated in Germany and it has a total length of 30 m, with six identical frames at 5 m spacing, a 17.5 m wide and height at eaves of 6 m. The total weight of the steel structure is 27716.3 kg + 2550.4 kg (hot rolled steel profiles + cold formed purlins), from which, all 27716.3 kg represent reused steel. Practically, the entire superstructure was considered to be reused in this case scenario. As in Case 2, the beams are IPE 360 steel profile and the columns are HEA 400.

The steel grade is S275. Only the purlins of the steel structure represent new steel in superstructure. The dimensions of the steel structure are the same as in Case 0 and in the Case 2.

The envelope consists in 80 mm steel sandwich panels with mineral wool insulation (reused elements) plus 60 mm mineral wool sandwich panels (new elements) to respect with the U-values existing in Case 0.

### 3. System boundaries

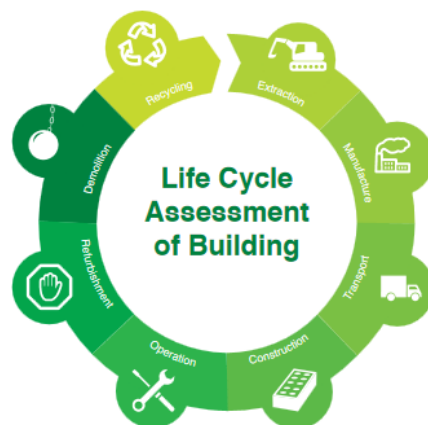
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The assessment was carried out having the following system boundaries:

- The heated floor area of the industrial hall is 525 m<sup>2</sup>;
- Other materials and components considered:
  - concrete foundations and concrete floor: 185 m<sup>3</sup>;
  - triple glazed windows: 22.5 m<sup>2</sup>;
  - sectional sliding gates: 48 m<sup>2</sup>;
- The U-value for the external walls is 0.56 W/m<sup>2</sup>K, for roof elements is 0.34 W/m<sup>2</sup>K, for ground floor slab is 0.76 W/m<sup>2</sup>K while for windows and entrance-door is 1.3 W/m<sup>2</sup>K;
- The operational lifetime of the building is 25 years;
- The assessment considers the main following building components: foundation and ground floor slab (concrete and steel rebars), steel load bearing structure (hot rolled and cold-formed steel elements), sandwich panels (steel sandwich panels with mineral wool insulation), triple glazed windows and sectional sliding gates.
- The steel rebars were counted as new material, with an input of 20% steel scrap in the process of manufacturing and an end-of-life scenario with 85% recycling potential and 15% landfilling or loss material after sorting [23].

## 4. Environmental assessment

Life cycle assessment method chosen for the evaluation of the environmental impact follows the rules of ISO 14044 [22], EN 15804 [22] and EN 15978 [22]. The approach for accounting of recycling in LCA is based on the modular building life cycle approach, where the benefits beyond the system boundary are reported separately (CEN, 2013). Global warming potential (GWP) was considered the indicator for the environmental performance.



*Fig. 15: Life Cycle Assessment representation [22]*

The life cycle assessment includes the following stages: “Product stage” (with modules A1 - Raw materials extraction, processing of secondary material input, A2 – Transport to the manufacturer, A3 – Manufacturing), “Construction stage” (with modules A4 – Transport to the building site, A5 - Building construction), “Use stage” (with module B6 - Operational energy use), “End of life” (C1 Deconstruction, demolition, C2 - transport to waste processing, C3 – waste processing for reuse, recovery and/or recycling, C4 - disposal) and “Benefits and loads beyond the system boundary” (module D – reuse and/or recycling potentials, expressed as net impacts and benefits) [22].

### 4.1 Lifecycle encounter

#### 4.1.1 Production stage: Modules A1-A3

This assessment takes into account the manufacture of the load bearing structure, foundations, floor slab, envelope, windows and sliding gates. LCA results are calculated for each case scenario considering a “new structure with new materials” and “reused structures” in which different amounts of reused steel from deconstructed industrial building halls were evaluated.

Product stage for the reused elements includes sandblasting and cutting welds (where required) 10 kgCO<sub>2</sub>e per ton of steel [22] and repainting from 260 to 280 kgCO<sub>2</sub>e per ton of steel [22].

#### 4.1.2 Construction stage: Modules A4-A5

In these modules the assessment includes the transportation of all construction materials from the manufacturer to the building site, equipment transportation, workers transportation and the structure erection. As the building site’s location was considered to be near Timisoara (a major city in Romania), all distances from manufacturers to the building site are between 15 and 70 km. For the reused steel structure (case 1) and elements (case 2 and 3) the distance considered for the transport of reused steel and sandwich panels was 1200 km. Euro 4 and Euro 5 articulated lorry

transport (20-26 t) and Euro 4 concrete mixer transport (12-14 t) were assigned to construction material transportation from GaBi database [22] and 65-80% of the payload was considered in the evaluation; also a return journey of the empty trucks was taken into account in the assessment. The transportation of workers (7 workers) was treated in the study considering an Euro 5 Minibus. The construction erection included excavation of soil for foundations and floor slab, concreting and assembling of steel structure and envelope using a 10t auto-crane, forklifts, man-lifts, wheel loader, bulldozer, excavator, concrete pump [22].

#### 4.1.3 Use stage: Module B6

The use stage refers to the building lifetime which starts from the completion of building construction and it ends when the building is deconstructed or demolished. The lifetime of the industrial hall in each case studied in this assessment is 25 years and therefore, during this lifetime, no maintenance, repair, elements replacements or any refurbishment were considered. The operational energy consumption assessment was based on energy simulation made by E-pass tool [22] using the climate condition based on weather data 'Timisoara Climate Normals (1961-1990)' [22]. Default value used for the heating of industrial hall was 18 °C. Neither heat recovery nor mechanical cooling was considered in the assessment. The use stage was identical for each case scenario therefore both new and reused buildings have the same environmental impact during use stage.

#### 4.1.4 End-of-life: Modules C1-C2

In this paper the end-of-life scenario for each one from the four cases studied involved an instance for "demolition and recycling" and another one for "deconstruction and reuse" (Module C1). The deconstruction included dismantling of steel structure and envelope using a 10t auto-crane, forklifts, man-lifts, hydraulic breaker excavator and wheel loader [22], while the demolition included working with 10t auto-crane, bulldozer, hydraulic breaker excavator, hydraulic scrap shear and wheel loader. The deconstruction process of the steel structure follows the reverse assembling process, to which additional effort is added to preserve the integrity of the deconstructed components for reuse [22]. The supplementary effort was generated in the study as 1.5 workload multiplier for the amount of reused elements in the end-of-life.

Transportation of residual materials deriving from deconstruction and demolition to the waste processing or nearest salvage yard is included in C2 module. The waste processing covers handling for reuse, recycling and disposal. According to the World Steel Association (2017) steel recycling is included in Module A.

#### 4.1.5 Benefits and loads beyond the system boundary: Module D

Recovered materials, recovered structure or elements of the structure, recycling of steel scrap can generate earnings as benefits and savings. In this paper data for Module D is based on scenarios considering current practices and current rates of materials recycling. The calculation model used for the assessment of Module D is based on an innovative calculation model [22], compatible with the methodology of the EN 15804, Product Environmental Footprint (European Commission, 2018) [22] standard and the World Steel Association (2017) approach [22]. The calculated impact involving potential environmental credits or burdens generated by future lifecycles is based on the

impacts of the reuse and the recycling of the materials jointly operated. The calculation is based on the inputs and outputs of recycled and reused materials, on the impact of the virgin material production and the impact of theoretical pure recycling. The yield of the recycling process was considered 0.916 t of recycled steel per 1 t of steel scrap (as specified by World Steel Association, 2017) and the information about Modules A1-3 were based on Environmental Product Declarations (EPD's) by Ruukki Construction [22].

#### 4.2 Assessed scenarios for the environmental impact

Each case from the four cases studied covered a view for “demolition and recycling” and one for “deconstruction and reuse” in the end-of-life module (except for the Case 0+ which was considered for reuse only, as it included a design for deconstruction). In the assessment was assumed that when it comes of reusing construction materials, both steel and envelopes are reused materials, as described in following scenarios. The percentage of steel used in the description below refers to the steel utilized in the superstructure (main and secondary structure) only:

- *New steel – demolition and recycling (Case 0):* in the assessment was considered that the external steel scrap input includes 20% from manufacturing processing (both blast furnace and electric arc furnace are used in the fabrication processing) [22] and that 90% of the steel is recycled at the structure's end-of-life;
- *New steel – deconstruction and reuse (Case 0):* in the assessment was considered that the external steel scrap input includes 20% from manufacturing processing, 90% of the steel is reused in the structure's end-of-life and 90% of the remaining steel is recovered for recycling. Also, it was considered that 90% of the envelope is reused and from the remaining sandwich panels 90% of the steel sheets are recovered for recycling (3.45% of the total mass of the sandwich panels);
- *New steel, including design for deconstruction with deconstruction and reuse in the end-of-life (Case 0+):* in the assessment was considered that the external steel scrap input includes 20% from manufacturing processing and 100% of the steel derived from superstructure is reused at the structure's end-of-life. Also, it was considered that 90% of the envelope is reused and from the remaining sandwich panels 90% of the steel sheets are recovered for recycling;
- *Reused steel structure – demolition and recycling (Case 1):* 65.10% of the total steel weight in the superstructure (incl. purlins) represented reused steel and the remaining new steel has an external scrap input of 20%. At the structure's end-of-life 90% of the steel is recycled (including 90% of the steel sheets in sandwich panels);
- *Reused steel structure – deconstruction and reuse (Case 1):* 65.10% of the total steel weight of the structure (incl. purlins) represented reused steel and the remaining new steel has an external scrap input of 20% from manufacturing processing. At the structure's end-of-life 72% of the steel load bearing structure (representing steel frames only) is reused and 90% of the remaining steel is recovered for recycling (25% of the total steel mass). Also, it was considered that 90% of the envelope is reused and from the remaining sandwich panels 90% of the steel sheets are recovered for recycling (3.45% of the total mass of the sandwich panels);
- *Reused steel elements – demolition and recycling (Case 2):* 55.49% of the total steel weight in superstructure (incl. purlins) represented reused steel and the remaining new steel has an external scrap input of 20%. At the structure's end-of-life 90% of the steel is recycled (including 90% of the steel sheets in sandwich panels);
- *Reused steel elements – deconstruction and reuse (Case 2):* 55.49% of the total steel weight of the structure (incl. purlins) represented reused steel and the remaining new steel has an external scrap input of 20%. At the structure's end-of-life 55.49% of the steel load bearing structure (representing columns and beams only) is reused and 90% of the remaining steel is recovered for recycling (40.06% of the total steel mass of the superstructure). Also, it was

considered that 90% of the envelope is reused and from the remaining sandwich panels 90% of the steel sheets are recovered for recycling;

*Table 5: Assessed scenarios for steel in superstructure (incl. purlins) – input and output flow of the material*

| End-of-life for steel in superstructure (recycle scenario) | In           |                 |                           | Out   |                    |                                |
|------------------------------------------------------------|--------------|-----------------|---------------------------|-------|--------------------|--------------------------------|
|                                                            | New material | Reused material | Recycled material (scrap) | Waste | Material for reuse | Material for recycling (scrap) |
| <b>Case 0</b>                                              | 80%          | 0%              | 20%                       | 10%   | 0%                 | 90%                            |
| <b>Case 1</b>                                              | 27.92%       | 65.10%          | 6.98%                     | 10%   | 0%                 | 90%                            |
| <b>Case 2</b>                                              | 35.61%       | 55.49%          | 8.90%                     | 10%   | 0%                 | 90%                            |
| <b>Case 3</b>                                              | 29.67%       | 62.91%          | 7.42%                     | 10%   | 0%                 | 90%                            |
| <b>Case 4</b>                                              | 0%           | 100%            | 0%                        | 0%    | 0%                 | 100%                           |
| End-of-life for steel in superstructure (reuse scenario)   | In           |                 |                           | Out   |                    |                                |
|                                                            | New material | Reused material | Recycled material (scrap) | Waste | Reused material    | Material for recycling (scrap) |
| <b>Case 0</b>                                              | 80%          | 0%              | 20%                       | 1%    | 90%                | 9%                             |
| <b>Case 0+</b>                                             | 80%          | 0%              | 20%                       | 0%    | 100%               | 0%                             |
| <b>Case 1</b>                                              | 27.92%       | 65.10%          | 6.98%                     | 3%    | 72%                | 25%                            |
| <b>Case 2</b>                                              | 35.61%       | 55.49%          | 8.90%                     | 4.45% | 55.49%             | 40.06%                         |
| <b>Case 3</b>                                              | 29.67%       | 62.91%          | 7.42%                     | 3.71% | 62.91%             | 33.40%                         |
| <b>Case 4</b>                                              | 0%           | 100%            | 0%                        | 0%    | 100%               | 0%                             |

- *Reused steel elements – demolition and recycling (Case 3):* 62.91% of the total steel weight of the structure (incl. purlins) represented reused steel and the remaining new steel has an external scrap input of 20%. At the structure's end-of-life it is assumed that 90% of the steel is recycled (including 90% of the steel sheets in sandwich panels which represents 34.4% of the total mass of the sandwich panels);
- *Reused steel elements – deconstruction and reuse (Case 3):* 62.91% of the total steel weight of the structure represented reused steel and the remaining new steel has an external scrap input of 20%. At the structure's end-of-life 62.91% of the steel load bearing structure (representing columns and beams only) is reused and 90% of the remaining steel is recovered for recycling. Also, it was considered that 90% of the envelope is reused and from the remaining sandwich panels 90% of the steel sheets are recovered for recycling.
- *Reused structure – demolition and recycling (Case 4):* 100% of the total steel weight in superstructure (incl. purlins) represented reused steel, only the rebars in foundations were considered new steel, with an input scrap of 20%. At the structure's end-of-life 100% of the steel is recycled (including 100% of the steel sheets in sandwich panels);
- *Reused structure – deconstruction and reuse (Case 4):* 100% of the total steel weight in superstructure (incl. purlins) represented reused steel. At the structure's end-of-life all the steel in the load bearing structure is reused;

#### 4.3 LCA results of the assessed cases

The environmental impact of the assessment was expressed using GWP as a pointer. Table 6 outlines the greenhouse gas (GHG) emissions for each studied case. As the results show, in the production stage (A1-A3) the savings are between 29-33% when reclaimed steel is used in construction compared with building using new materials only. Relocating the entire superstructure

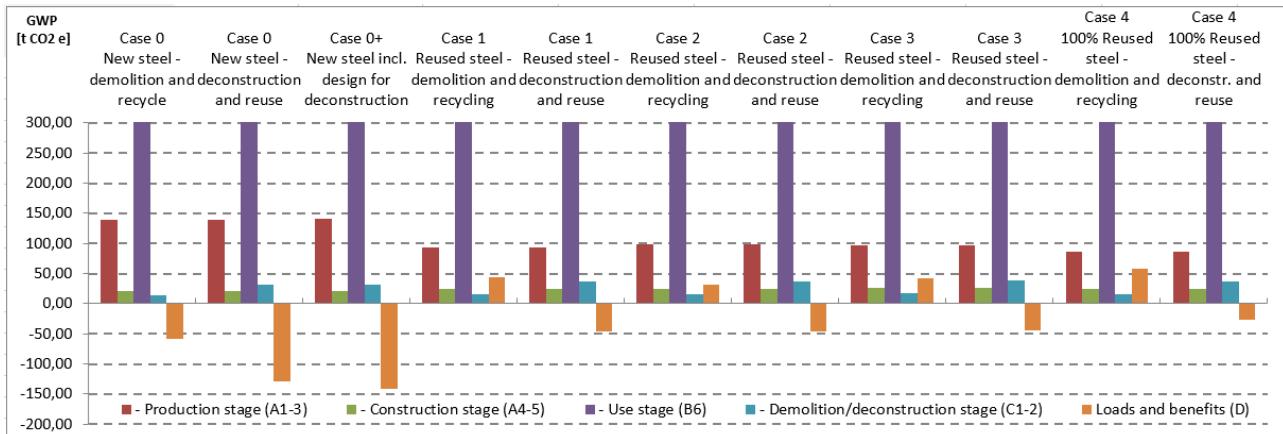
*Table 6: LCA results of the assumed scenarios*

| Scenario                                  | Demolition and recycle                                            | Deconstruction and reuse | Demolition and recycle                     | Deconstruction and reuse | Demolition and recycle                     | Deconstruction and reuse |
|-------------------------------------------|-------------------------------------------------------------------|--------------------------|--------------------------------------------|--------------------------|--------------------------------------------|--------------------------|
|                                           | Reused steel (case 1) [tCO <sub>2</sub> e]                        |                          | Reused steel (case 2) [tCO <sub>2</sub> e] |                          | Reused steel (case 3) [tCO <sub>2</sub> e] |                          |
| <b>GWP impact (A-C)</b>                   | <b>591.90</b>                                                     | <b>613.59</b>            | <b>600.60</b>                              | <b>620.79</b>            | <b>600.79</b>                              | <b>622,05</b>            |
| -Production stage (A1-3)                  | 92.47                                                             | 92.47                    | 98.97                                      | 98.97                    | 97.20                                      | 97.20                    |
| -Construction stage (A4-5)                | 23.72                                                             | 23.72                    | 24.42                                      | 24.42                    | 25.63                                      | 25.63                    |
| -Use stage (B6)                           | 460.77                                                            | 460.77                   | 460.77                                     | 460.77                   | 460.77                                     | 460.77                   |
| -Demolition / deconstruction stage (C1-2) | 14.94                                                             | 36.63                    | 16.44                                      | 36.63                    | 17.19                                      | 38.45                    |
| Loads and benefits (D)                    | 43.12                                                             | -45.75                   | 31.37                                      | -46.18                   | 42.63                                      | -43.72                   |
| Scenario                                  | 100% Reused steel in superstructure (case 4) [tCO <sub>2</sub> e] |                          | New steel (case 0) [tCO <sub>2</sub> e]    |                          | New steel (case 0+) [tCO <sub>2</sub> e]   |                          |
| <b>GWP impact (A-C)</b>                   | <b>586.97</b>                                                     | <b>607.07</b>            | <b>634.78</b>                              | <b>651.79</b>            | <b>654.97</b>                              |                          |
| -Production stage (A1-3)                  | 85.40                                                             | 85.10                    | 139.79                                     | 139.79                   | 141.17                                     |                          |
| -Construction stage (A4-5)                | 24.36                                                             | 24.36                    | 20.49                                      | 20.49                    | 21.21                                      |                          |
| -Use stage (B6)                           | 460.77                                                            | 460.77                   | 460.77                                     | 460.77                   | 460.77                                     |                          |
| -Demolition / deconstruction stage (C1-2) | 16.44                                                             | 36.54                    | 13.73                                      | 30.74                    | 31.82                                      |                          |
| Loads and benefits (D)                    | 58.10                                                             | -26.88                   | -58.39                                     | -128.44                  | -141.57                                    |                          |

of the industrial hall and reusing as it is, involves a decrease of the emissions in the production stage of up to 40%. Including the design for deconstruction in the baseline scenario and the additional amount of steel which it implies, lifts the emissions in the production stage with 1% in

comparison with the optimal design scenario of the industrial hall. In the end-of-life stage (C1-C2), when the process of deconstruction requires additional effort to preserve the integrity of the deconstructed components for reuse, the savings are acquired by the recycling scenarios.

According to the results, highest environmental impact is shown by the cases when structures erected with new elements are deconstructed for the next reuse case (1.24 t CO<sub>2</sub> e/m<sup>2</sup> – case 0 and 0+) while the lowest rate of emissions are registered when structures are built with reused portal frames and at the end-of-life of the structure steel is recovered for recycling (0.97 t CO<sub>2</sub> e/m<sup>2</sup> – case 1).

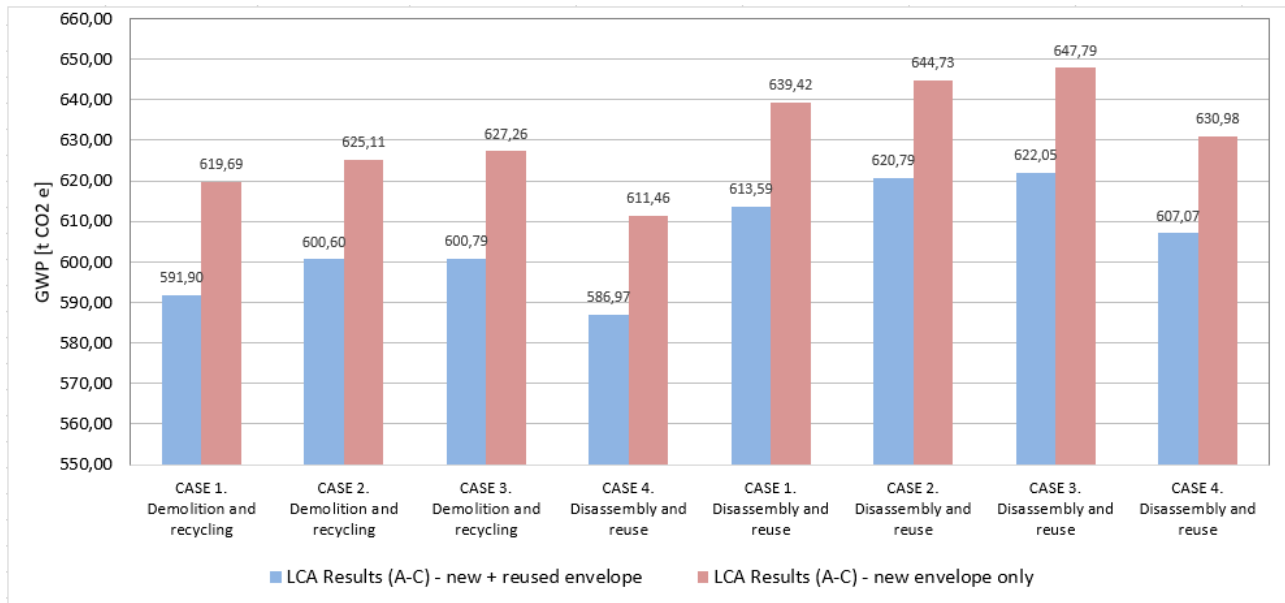


*Fig. 16: Total LCA results of the scenarios including loads and benefits beyond the system boundary*

Figure 16 presents the total LCA results of the assessed scenarios of an industrial hall. The LCA savings are reflected as negative values while positive values define burdens of material utilisation. It can be seen that benefits and loads beyond the system boundary are not aggregated with the life cycle impacts (Modules A to C), as provided by the CEN/TC 350 methodology.

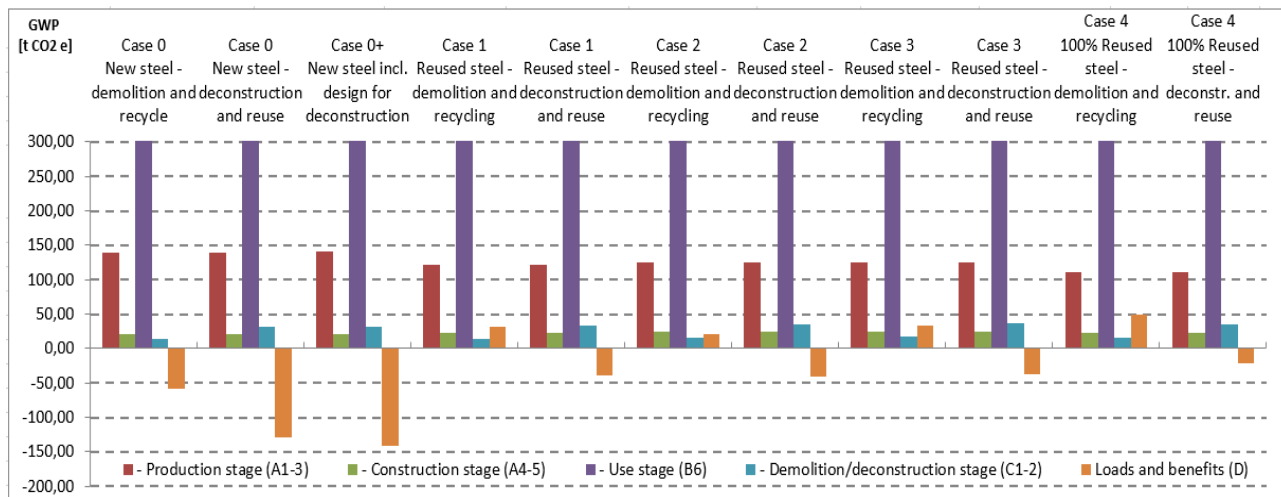
The results show that beyond the system boundary, in the scenarios where the structures built with reused elements and at the end-of-life steel is recovered for recycling, a burden is recorded in the assessment. The highest potential savings (0.244-0.269 t CO<sub>2</sub> e/m<sup>2</sup>) appear in the scenario when the industrial hall was erected with new elements which are deconstructed for the next reuse case in the end-of-life.

When in the construction of an industrial hall only the steel in main structure is reused (in different amounts) and all the other materials, including materials for the envelope, are new materials, the emissions in the production stage increase with 26-32% (in comparison with the scenarios where the structure is built using reclaimed elements for both load bearing structure and envelope). As showed in figure 17, the total emissions rise with 0,045 t CO<sub>2</sub> e/m<sup>2</sup> when the envelope consists only of new sandwich panels. In the scenario where the sandwich panels were also reused, a second layer of sandwich panels was required, to comply with the U-values, which drove to additional energy consumption but a significant decrease of emissions in the production stage 0,049 t CO<sub>2</sub> e/m<sup>2</sup>).



*Fig. 17: LCA results for envelope consisting of both new and reused sandwich panels (in blue) vs. new envelope only (in red)*

The total LCA results of the assessed scenarios of the industrial hall, when the envelope consists in new sandwich panels for all scenarios (both structures made with new steel and those built with reused steel) are presented in figure 18.



*Fig. 18: Total LCA results of the scenarios including loads and benefits beyond the system boundary (when envelope in all cases consists in new sandwich panels)*

As shown in the figure 18, using only new sandwich panels for the envelope in cases 1, 2, 3 and 4 increase the emissions in the production stage and induced a cut down of CO<sub>2</sub>e in the construction and demolition/deconstruction stage. A comparative LCA for cases 0 and 4 (when the structure is build either with new envelope or with new + reused envelope) shows a sharp difference between the stages in each case scenario and the impact of reusing only new envelope, as it can be seen in figure 19.

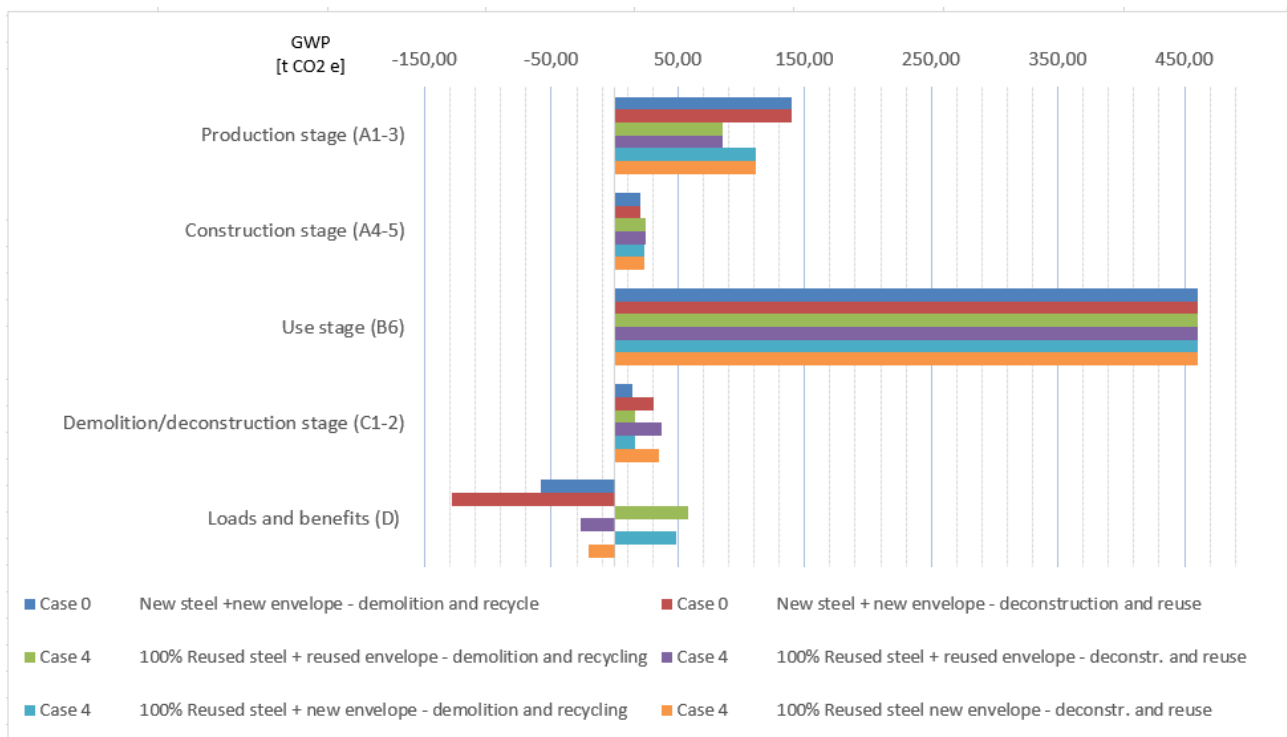


Fig. 19: Comparative LCA results for cases 0 and 4 (when envelope consists in new sandwich panels or new + reused sandwich panels)

## 5. Economic assessment

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The calculation of economic indicators is based on the same scenarios and modules as in the LCA analysis, referring to new steel and reused steel structures, to which were associated time and financial costs. The assessment of the economic performance of studied cases follows the rules described in EN 16627 [1].

### 5.1 Assessed scenarios for the economic impact

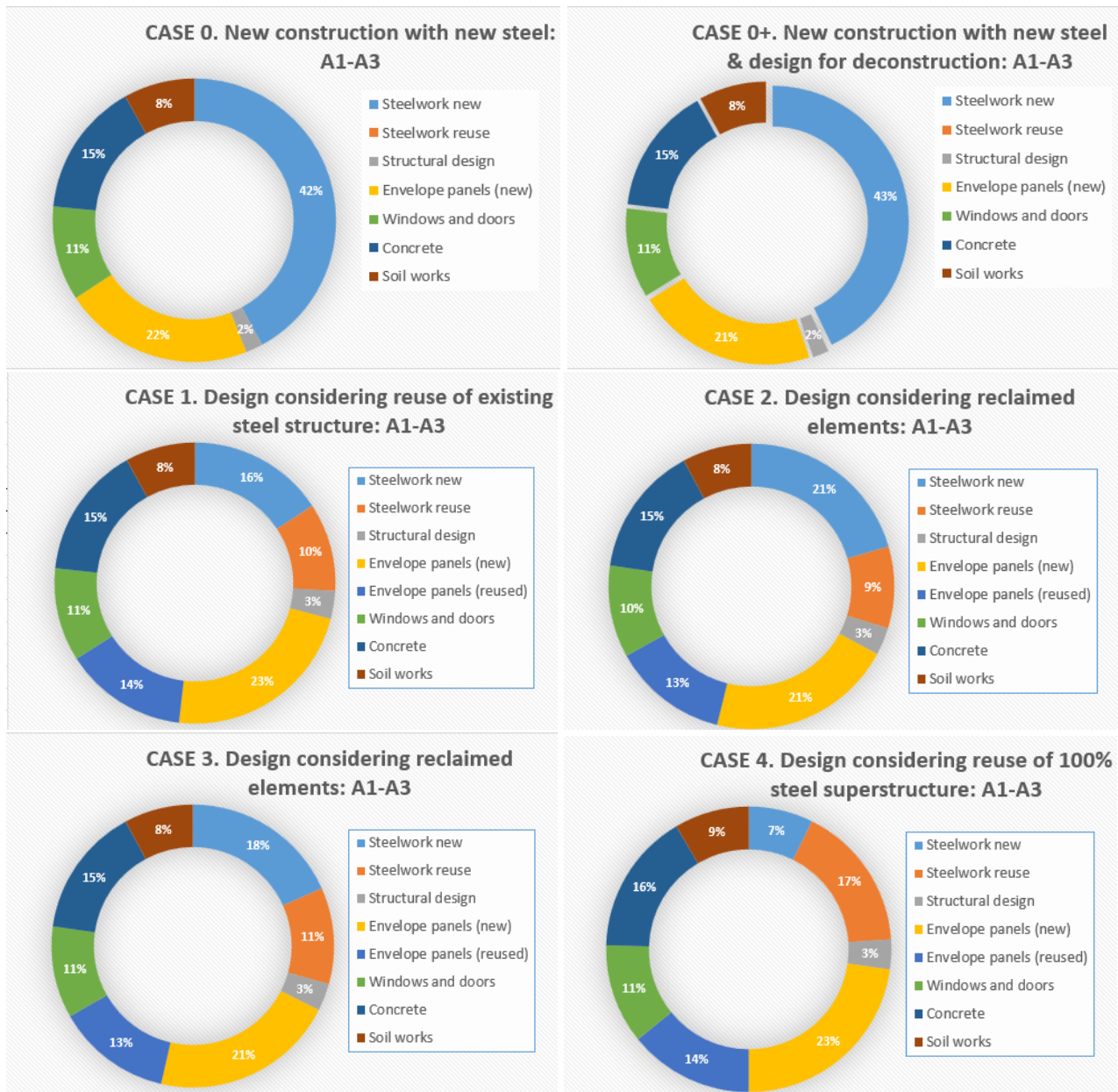
Potential cost savings includes revenues for recycled steel (earnings from the sold steel scrap), revenues from the sold steel elements or sold structure and revenues from the sold sandwich panels. The returns considered in case of recycling were of 180 euro/ton of sold steel scrap (the value represents the price of auctioned steel scrap in Romania at the moment of the assessment), while the gains from the sold recovered structure (without refurbishment) were calculated at a value of 639 euro/ton of steel. Additional costs for expertise, redesign, testing, sandblasting, repainting were counted in cases where reused steel was involved [22].

### 5.2 LCC results of the assessed cases

The results of the life cycle cost assessment show that the economic impact in the production stage (Modules A1-A3) is represented by a 0.5-3% increase of the cost when construction materials are reused. This is due to the fact that reused materials need additional expertise, testing, redesign and/or sandblasting. However, when the entire (100%) superstructure is relocated and reused, the economic impact in the production stage is smaller than the economic impact of the structures build with new materials.

Figure 20 shows the economic contribution of each construction material in the production phase: when using new materials only, the preponderant cost (42-43%) came from the cost of the steel, while in the reuse scenarios (cases 1, 2, 3 and 4) the highest costs come from the production of the envelope.

The higher economic impact of the scenarios which implicate reused materials is maintained also in the assessment of the total life cycle, when the costs increase with 10 to 22 euro/m<sup>2</sup> compared to the scenarios which relate to new materials only. Total life cycle cost considers also the costs of demolition, deconstruction and transportation of dismantled materials, which is why the savings gained in the production stage related with the reusing of the construction materials are counterbalanced by the costs of the deconstruction of the structure in the end-of-life stage.



*Fig. 20: The economic contribution of each construction material used during the production stage*

Life cycle costs results of the assessment for all life cycle stages, including potential savings, are given in Table 7 (potential savings are reflected as negative values). According to the results, highest LCC (modules A-C) is shown by the cases when structures erected with reclaimed elements are deconstructed for the next reuse case (608 €/m<sup>2</sup>) and lowest LCC by the case when construction uses new steel structures and after demolition recovered steel is sold for recycling (547 €/m<sup>2</sup>).

Table 7: Total LCC results of the assumed scenarios

| Scenario                                         | Reused steel<br>(case 1) [k euro]                           |                                  | Reused steel<br>(case 2) [k euro] |                                  | Reused steel<br>(case 3) [k euro] |                                  |
|--------------------------------------------------|-------------------------------------------------------------|----------------------------------|-----------------------------------|----------------------------------|-----------------------------------|----------------------------------|
|                                                  | Demolition<br>and recycle                                   | Deconstruc-<br>tion and<br>reuse | Demolition<br>and recycle         | Deconstruc-<br>tion and<br>reuse | Demolition<br>and recycle         | Deconstruc-<br>tion and<br>reuse |
| <b>Economic impacts (A-C)</b>                    | <b>297.89</b>                                               | <b>315.07</b>                    | <b>300.43</b>                     | <b>318.87</b>                    | <b>300.41</b>                     | <b>319.36</b>                    |
| -Production stage (A1-3)                         | 94.98                                                       | 94.98                            | 97.79                             | 97.79                            | 97.23                             | 97.23                            |
| -Construction stage (A4-5)                       | 44.61                                                       | 44.61                            | 44.25                             | 44.25                            | 44.67                             | 44.67                            |
| -Use stage (B6)                                  | 146.00                                                      | 146.00                           | 146.00                            | 146.00                           | 146.00                            | 146.00                           |
| -Demolition / deconstruction stage (C1-2)        | 12.31                                                       | 30.05                            | 12.39                             | 30.83                            | 12.51                             | 31.46                            |
| Loads and benefits (D)                           | -7.37                                                       | -16.83                           | -7.82                             | -18.22                           | -8.05                             | -18.93                           |
| <b>Total costs (including potential savings)</b> | <b>290.52</b>                                               | <b>298.81</b>                    | <b>292.61</b>                     | <b>300.65</b>                    | <b>292.36</b>                     | <b>300.43</b>                    |
| Scenario                                         | 100% Reused steel in<br>superstructure<br>(case 4) [k euro] |                                  | New steel<br>(case 0) [k euro]    |                                  | New steel<br>(case 0+) [k euro]   |                                  |
|                                                  | Demolition<br>and recycle                                   | Deconstruc-<br>tion and<br>reuse | Demolition<br>and recycle         | Deconstruc-<br>tion and<br>reuse | Deconstruction and reuse          |                                  |
| <b>Economic impacts (A-C)</b>                    | <b>294.18</b>                                               | <b>312.62</b>                    | <b>287.52</b>                     | <b>302.57</b>                    | <b>304.71</b>                     |                                  |
| -Production stage (A1-3)                         | 90.24                                                       | 90.24                            | 94.47                             | 94.47                            | 95.75                             |                                  |
| -Construction stage (A4-5)                       | 45.55                                                       | 45.55                            | 36.13                             | 36.13                            | 36.59                             |                                  |
| -Use stage (B6)                                  | 146.00                                                      | 146.00                           | 146.00                            | 146.00                           | 146.00                            |                                  |
| -Demolition / deconstruction stage (C1-2)        | 12.39                                                       | 30.83                            | 10.91                             | 25.96                            | 26.37                             |                                  |
| Loads and benefits (D)                           | -7.82                                                       | -18.22                           | -6.55                             | -18.71                           | -19.25                            |                                  |
| <b>Total costs (including potential savings)</b> | <b>286.36</b>                                               | <b>294.40</b>                    | <b>280.97</b>                     | <b>283.86</b>                    | <b>285.46</b>                     |                                  |

Figure 21 presents the total LCC results of the assessed scenarios, where the savings are reflected as negative values. When considering the loads and benefits from Module D only, the highest savings (34,70 euro/m<sup>2</sup> – 36,66 euro/m<sup>2</sup>) come when the structure is deconstructed for a future reuse in the end-of-life, but when the total life cycle is considered, the costs of the structure built with reused construction materials (cases 1, 2, 3 and 4) exceed the costs of the structure designed with elements from new materials (case 0 and 0+).

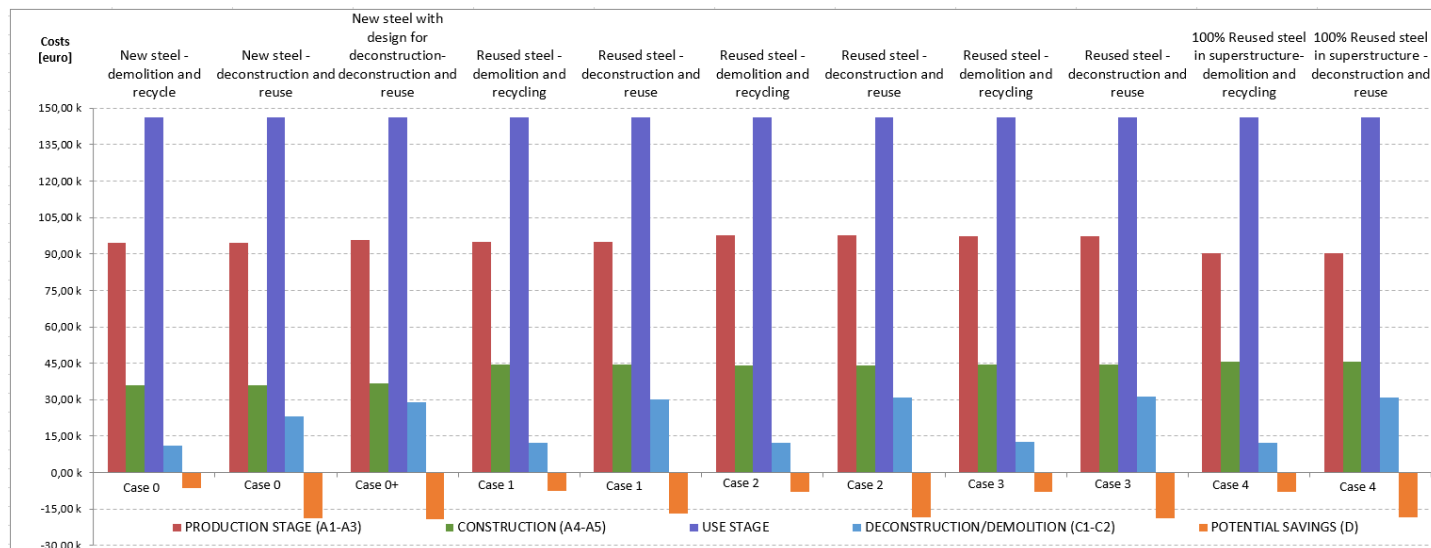


Fig. 21: LCC results of the scenarios, including potential savings

However, in the scenarios where the structure is built with reclaimed elements, the costs fall by 6-7.3% when the envelope consists only of new sandwich panels and only steel is reused. This leads to a change in the ranking (as showed in figure 21), where constructions built with reused steel elements surpass the structures erected with new steel profiles. In this case, the highest LCC, including potential savings, (543 €/m<sup>2</sup>) is shown by the scenario when construction consists of new steel elements and after demolition, steel scrap is sold for recycling – case 0+ and lowest LCC (modules A-D) by the case 4, when construction uses reclaimed steel and after demolition recovered steel is sold for recycling (508 €/m<sup>2</sup>).

The total LCC results of the assessed scenarios of the industrial hall, when the envelope consists in new sandwich panels for all scenarios (both structures made with new steel and those built with reused steel) are presented in figure 22.

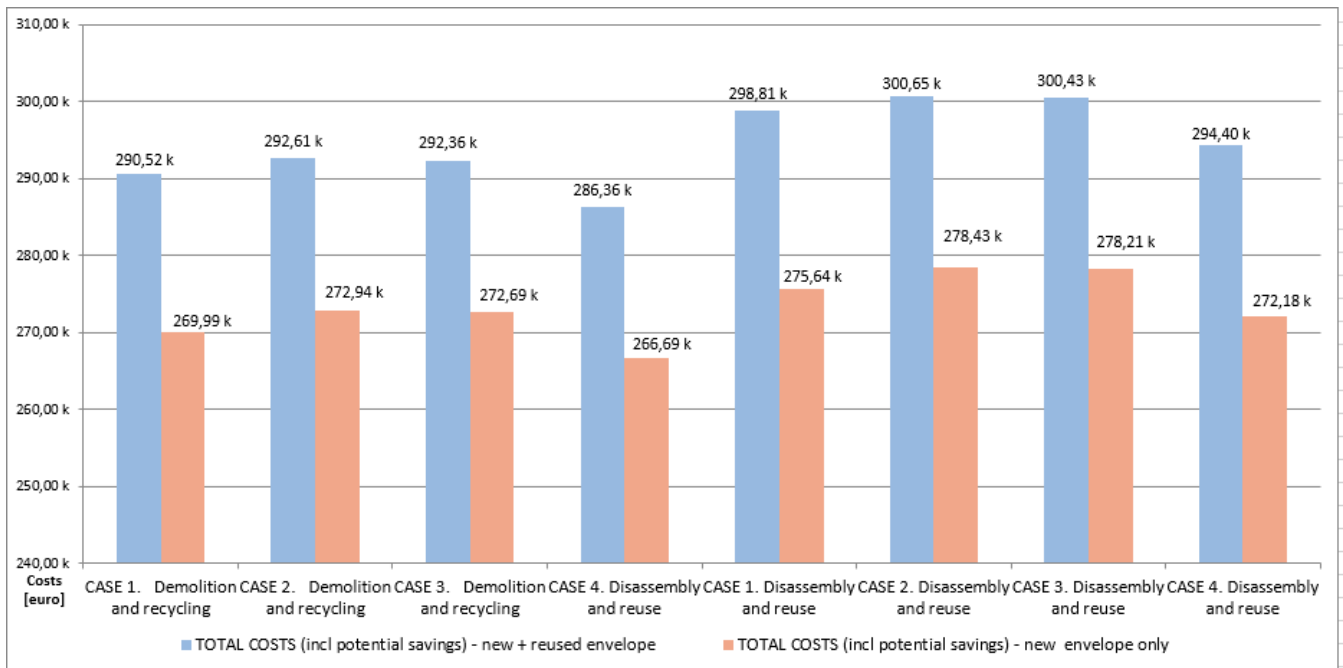


Fig. 22: LCC results for envelope consisting of both new and reused sandwich panels (in blue) vs. new envelope only (in red)

As shown in the figure 23, using only new sandwich panels for the envelope in cases 1, 2, 3 and 4 conducted to lower costs in the construction and demolition/deconstruction phase, and lower costs on the whole. In figure 24 it can be seen a comparative LCC assessment for cases 0 and 4 (when the structure is build either with new envelope or with new+reused envelope) which outlines a definite variation of the costs between the stages in each case scenario and the impact of reusing only new envelope.

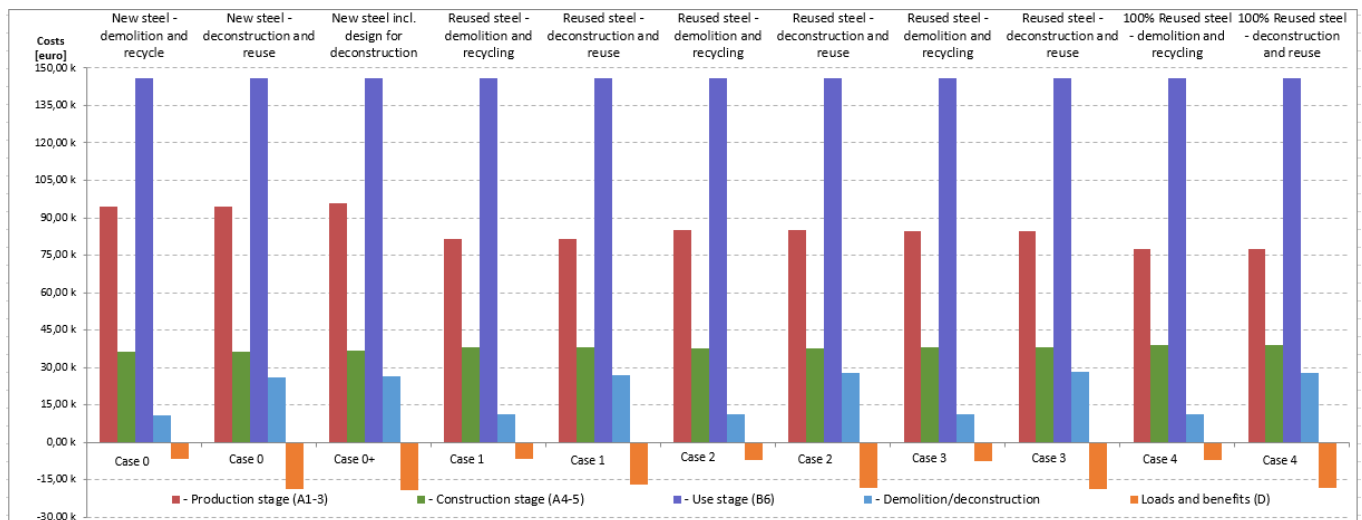
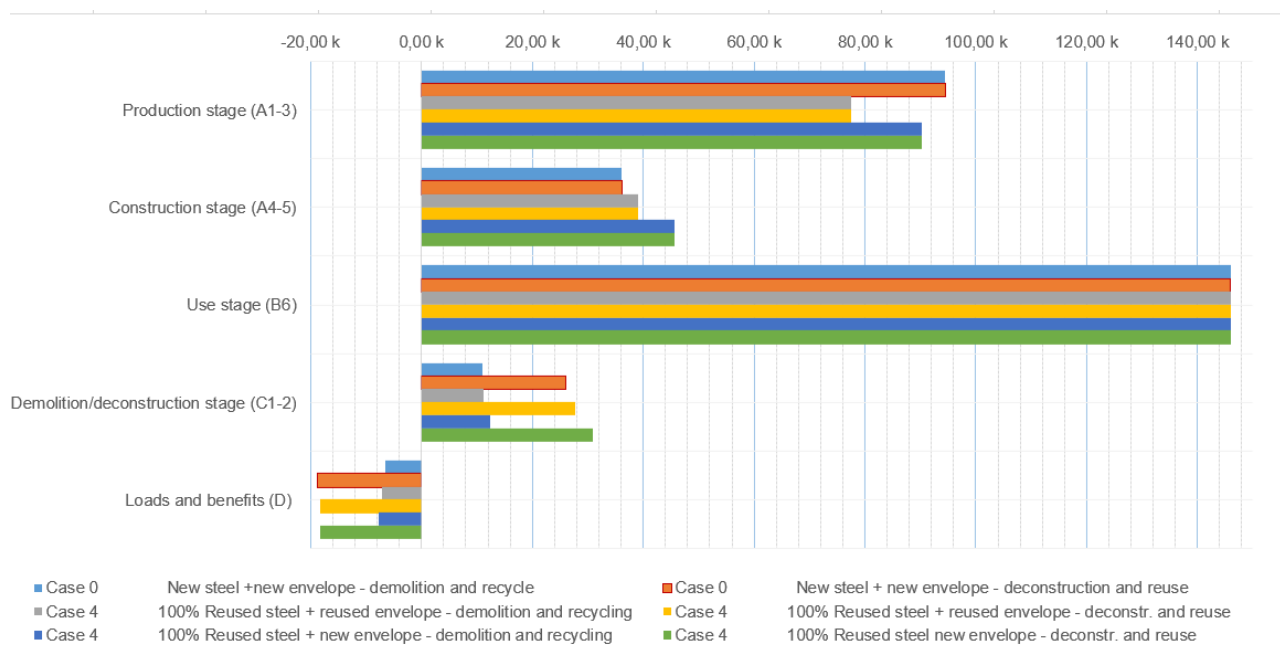


Fig. 23: Total LCC results of the scenarios including loads and benefits beyond the system boundary (when envelope in all cases consists in new sandwich panels)



*Fig. 24: Comparative LCC results for cases 0 and 4 (when envelope consists in new sandwich panels or new + reused sandwich panels)*

## 6. Summary and conclusions

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The end-of-life qualities of construction products play an important role, considering the circular economy, as buildings have a long intended life-span and require a significant amount of material resources and, by default, energy consumption. In order to maintain circularity, these resources have to be kept in-use through reuse, reclaim or recycling. The end-of-life credentials of structural materials varies notable due to their inherent properties and the final structural form, therefore the reuse of steel is a viable solution towards meeting circular economy criteria and lowering energy demand for the construction phase.

The present assessment considered six different study-cases, where the building is constructed from either new or reused components, and where the components will be either reused or recycled in the future in order to approach the costs and impacts in different life cycle stages beyond the system boundaries. The assessment stressed the structural feasibility, the environmental benefits and economic savings of a construction strategy based on the reuse approach.

The Life Cycle Assessment results showed that the reuse approach is a strategy that holds environmental benefits superior to recycling approach (modules A-C), the greatest gain being visible in the production stage (A1-3) where GHG emissions are between 29-33% smaller when the structure is built with reused steel (188.51 kg CO<sub>2</sub> e/m<sup>2</sup> for structures built with reused steel in comparison with 266.26 kg CO<sub>2</sub> e/m<sup>2</sup> for structures built with new steel).

According to the results, when it comes to economic potential of the reuse stages in various reuse and recycle scenarios, the scenario with reused steel elements resulted in higher potential savings (between 34.70 €/m<sup>2</sup> and 36.66 €/m<sup>2</sup>) compared to recycling scenario (between 12.47 €/m<sup>2</sup> and 15.33€/m<sup>2</sup>).

Another meaningful finding of the study case was that in cases when the structure is built with reclaimed elements for the main structure but the envelope consists only of new sandwich panels, the total costs fall by 6-7.3% in comparison with structures build with new materials only.

Nonetheless, due to the large variation of materials and building practices, LCC and LCA calculations are recommended to be carried out on a case-by-case basis, as the results could differ significantly from the present study. Some reasons for obtaining differences in results are the lack of significant quality data for deconstruction, the proximity of the new construction to the dismantled building, the burdens (energy consumption and costs) related to the transportation of new steel, depending on where steel is purchased from, the durability of materials and technical performance of envelopes are subject to chance, depending on the previous life span of the envelope and the fact that the value of reusable construction materials ranges from one country to another in the European market. Though, reuse of structural steel is a viable solution in construction industry, offering considerable benefits in terms of savings of embedded carbon in steel, of time savings or cost savings, when the interconnection between main actors involved in the reuse chain is closed in a strong loop.

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